AVISTA Model Report

An extensible desktop platform for tabular machine learning and deep learning analytics.

Project: national-poll2

Target: Number_of_Doctors_Visited

Original dataset rows: 714

Features: 13

Original split rows: 499 train / 72 validation / 143 test

Effective training rows: 541

Imbalance method: smote

Validation role: deep-model early stopping and checkpoint selection

Test role: held-out final evaluation after fitting and checkpoint selection

CV role: outer validation folds are used only for fold scoring

Cross-validation: 5 folds

Models trained: 5

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AVISTA version: 1.1.0

AVISTA release date: September 20, 2026

TabPFN 2.5 input: raw DataFrame with TabPFN-owned preprocessing

TabPFN external resampling: No

TabPFN available/effective training rows: 499/499

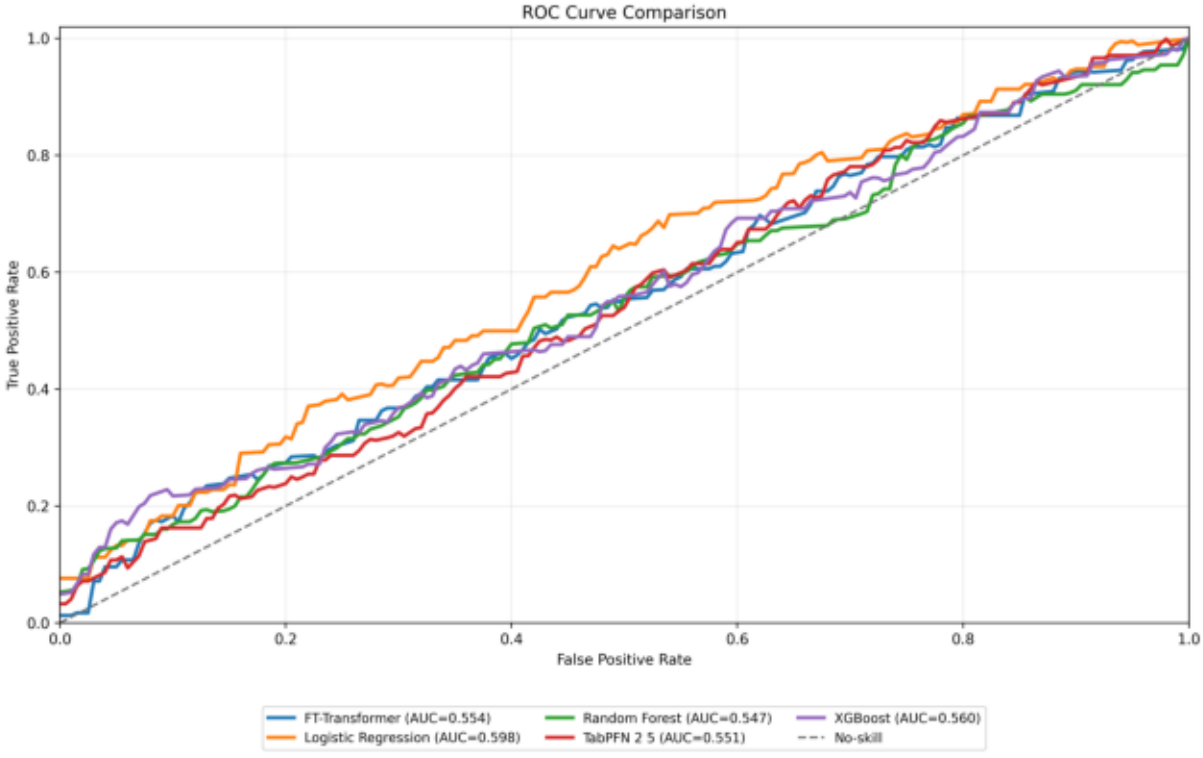
TabPFN supported row limit: 50000

Unless otherwise stated, model comparison figures and diagnostic tables are based on the test set.

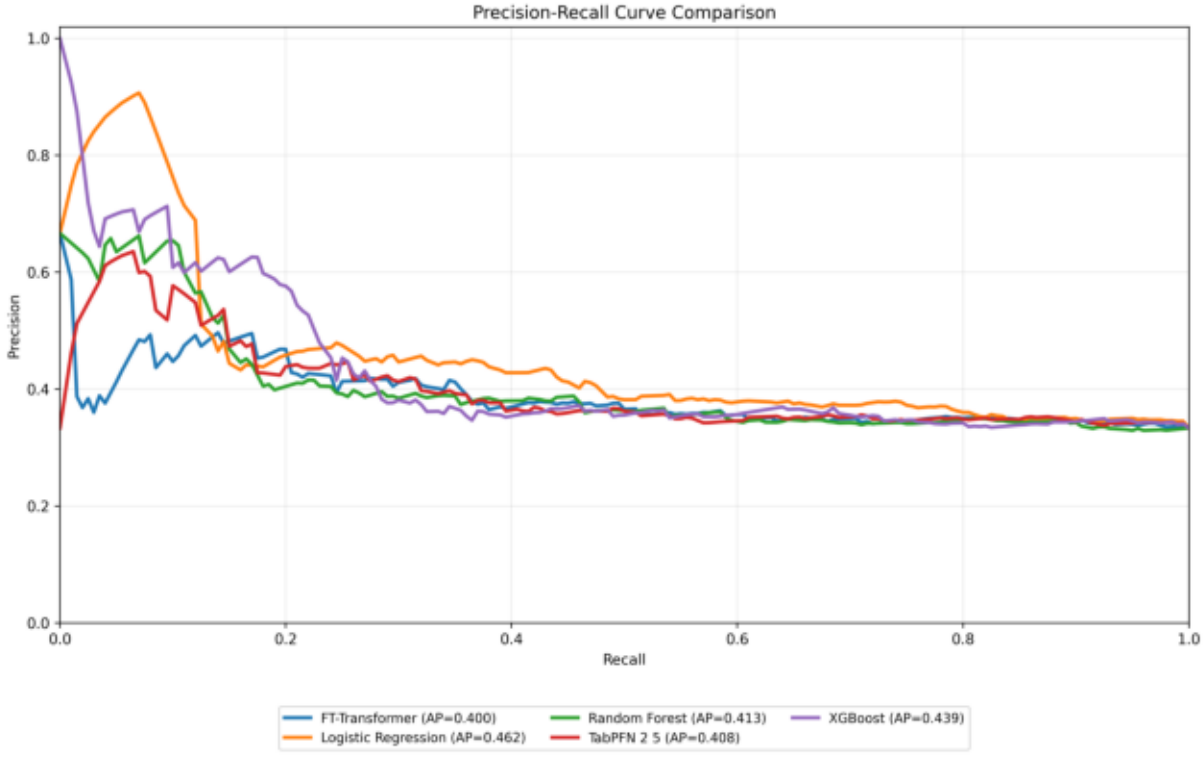
Model Performance Summary

Model	Status	Validation Accuracy	Validation Macro-F1	Test Accuracy	Test Macro-F1	ROC-AUC	Saved
Logistic Regression	trained	0.3611	0.3677	0.3986	0.3764	0.6002	Yes
Random Forest	trained	0.4028	0.3372	0.4336	0.3620	0.5456	Yes
XGBoost	trained	0.4167	0.2817	0.4755	0.3821	0.5589	Yes
FT-Transformer	trained	0.3750	0.3722	0.3357	0.3329	0.5536	Yes
TabPFN 2.5	trained	0.5278	0.2303	0.5175	0.2273	0.5483	Yes

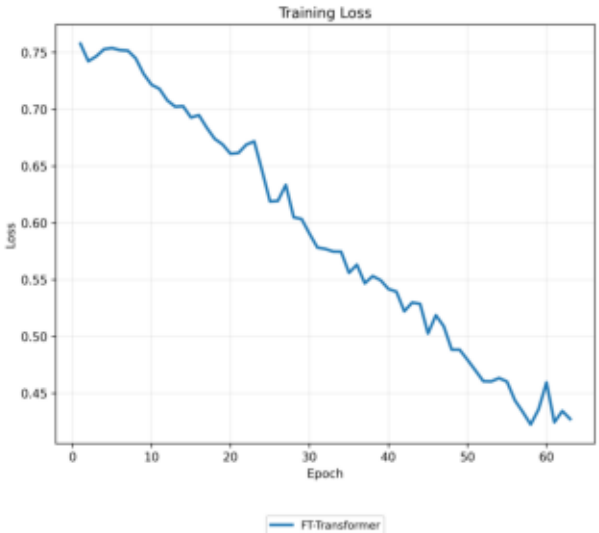
ROC Curve Comparison

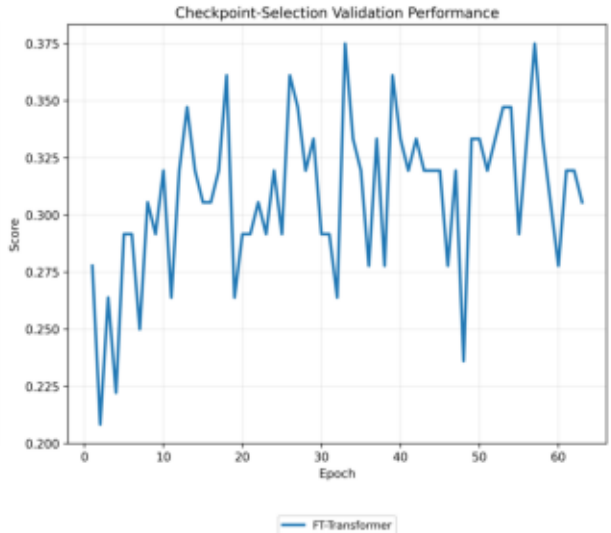


Precision-Recall Curve Comparison

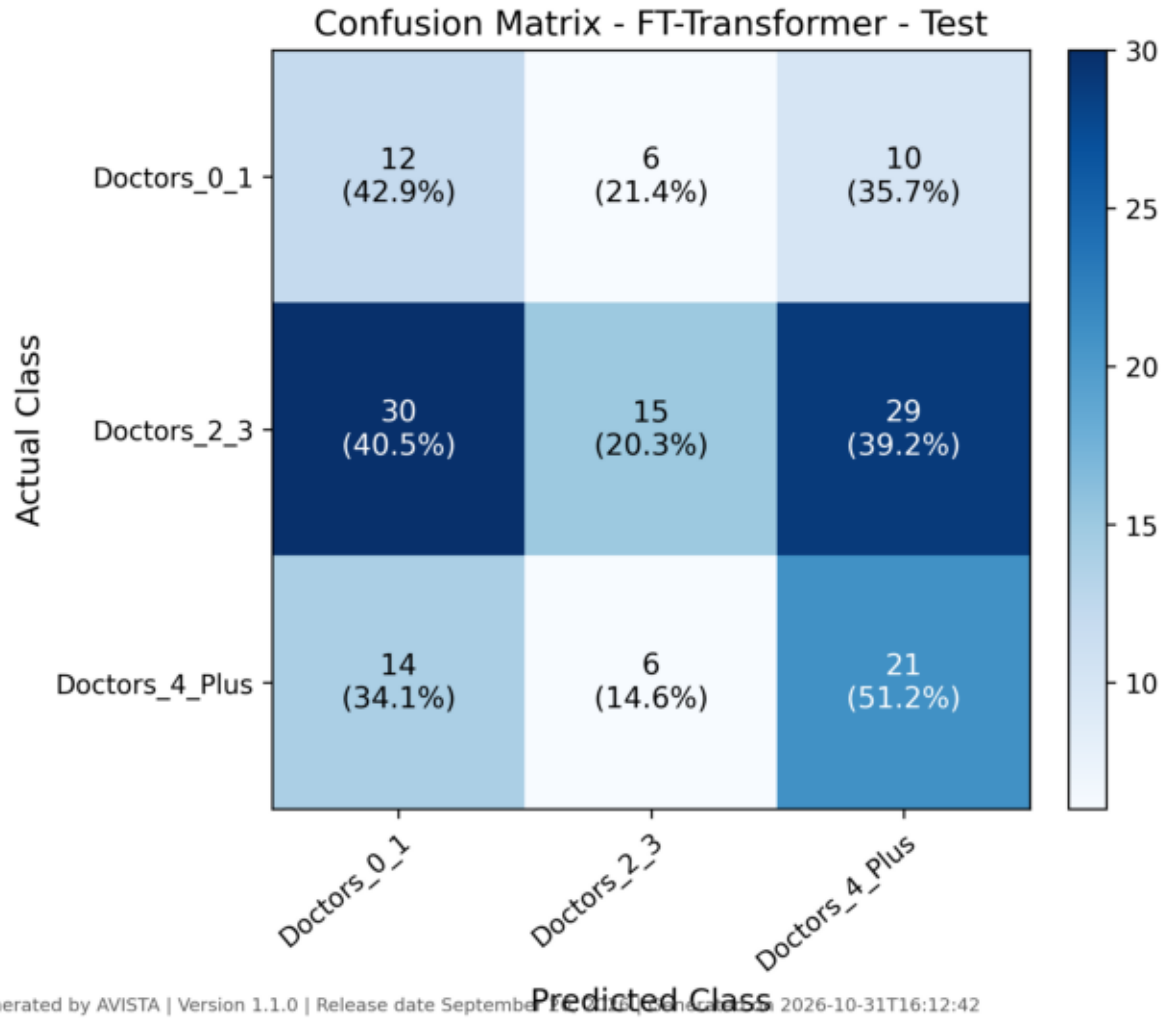


Deep Learning Training Curves



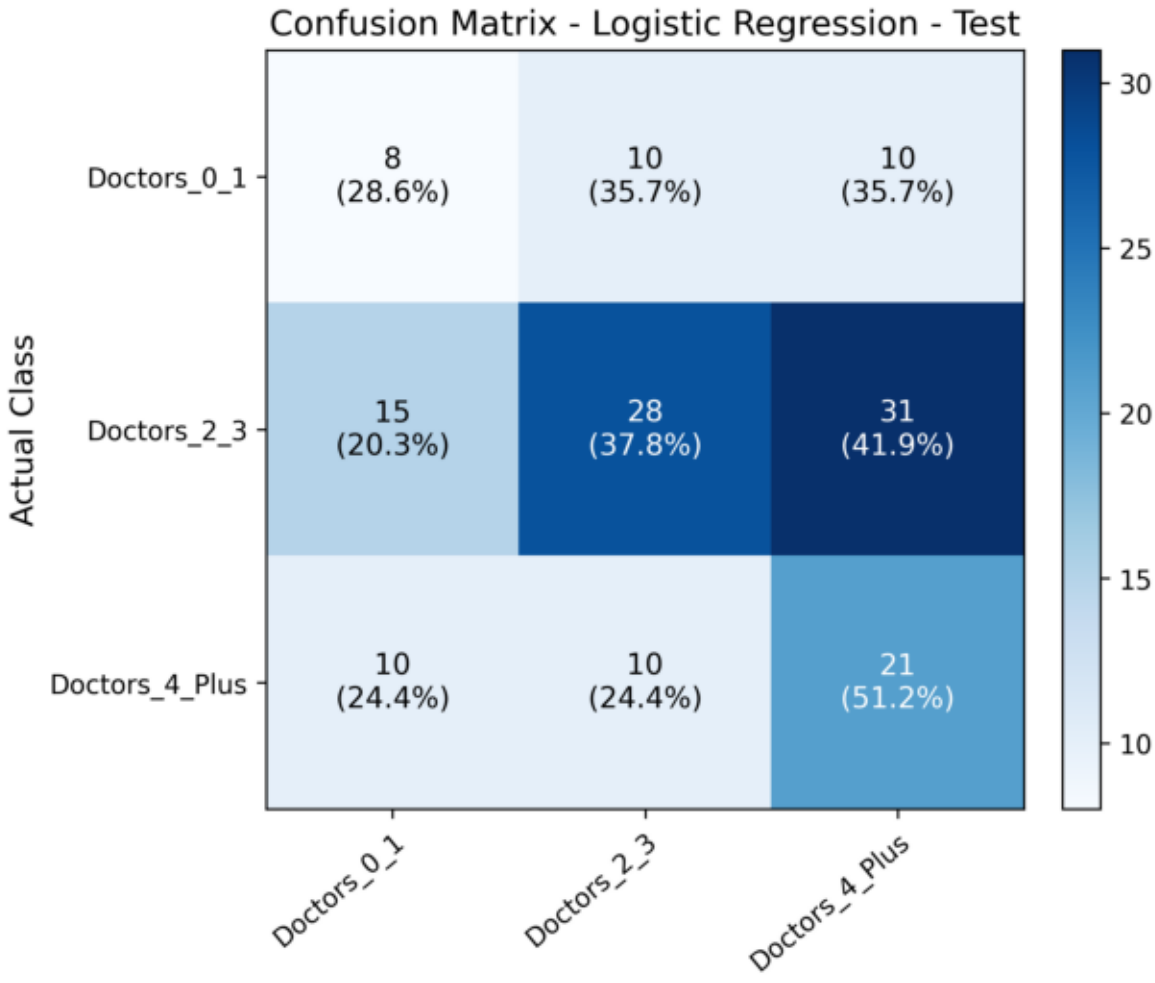


Confusion Matrices - FT-Transformer



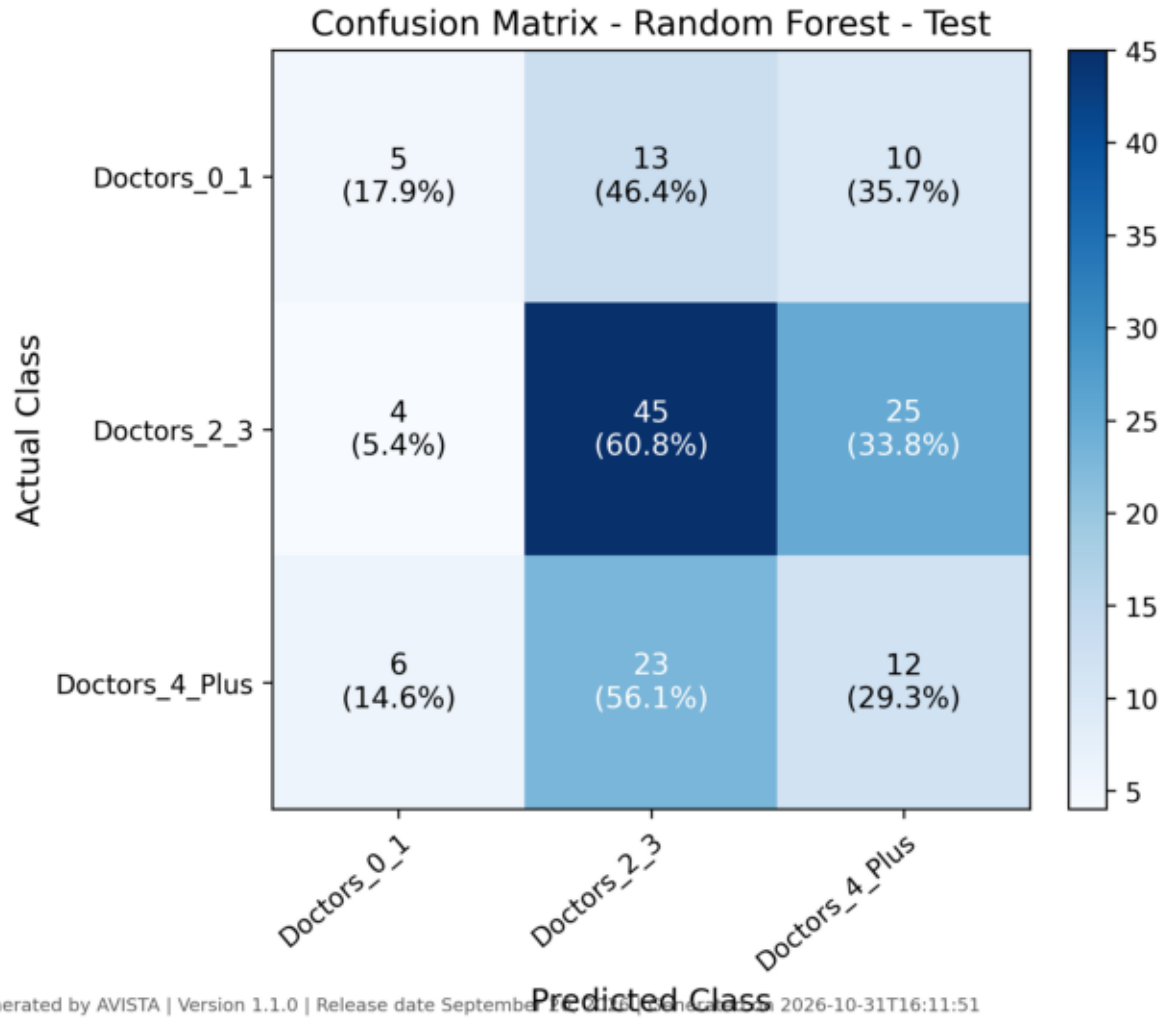
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Confusion Matrices - Logistic Regression



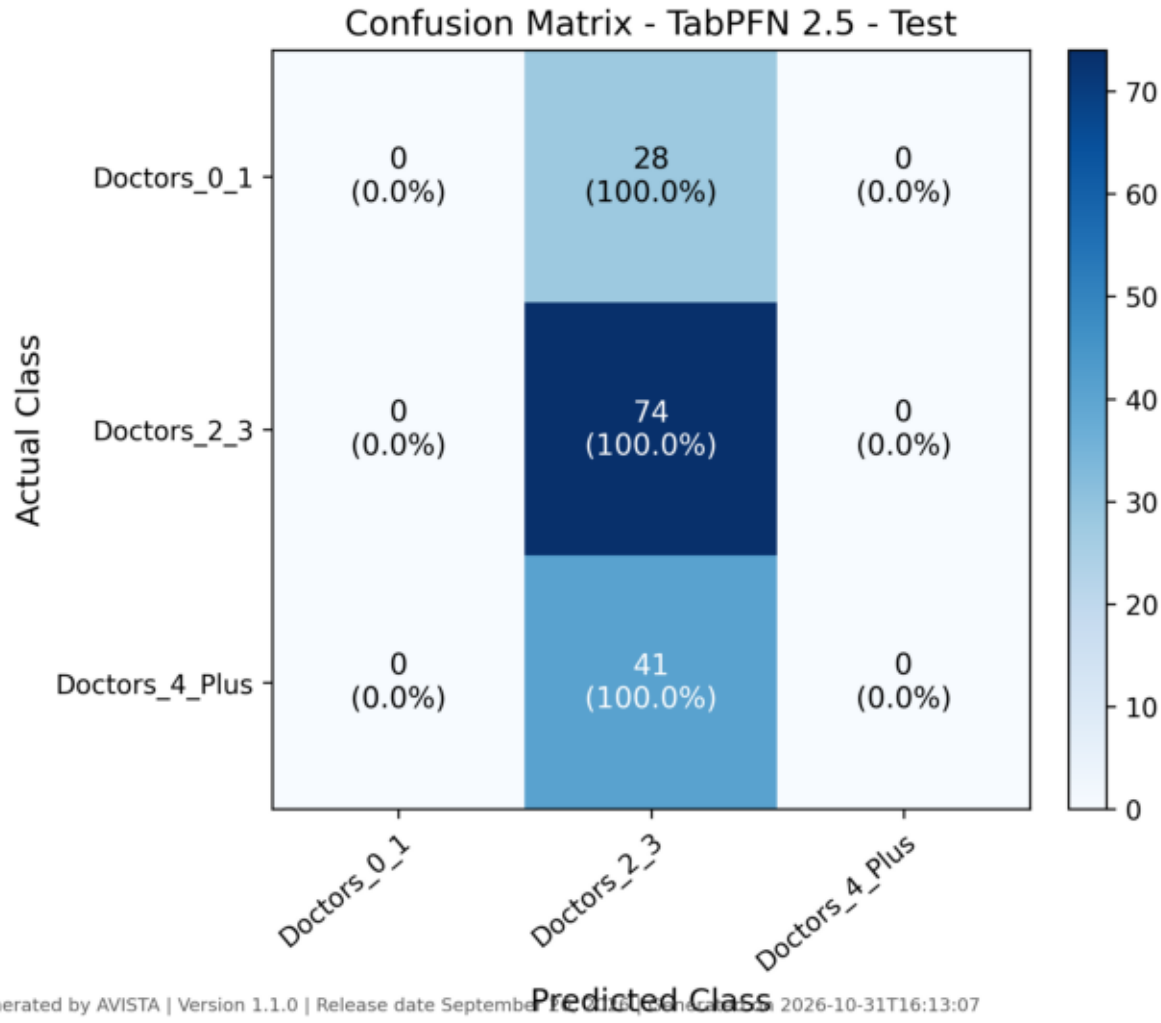
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Confusion Matrices - Random Forest



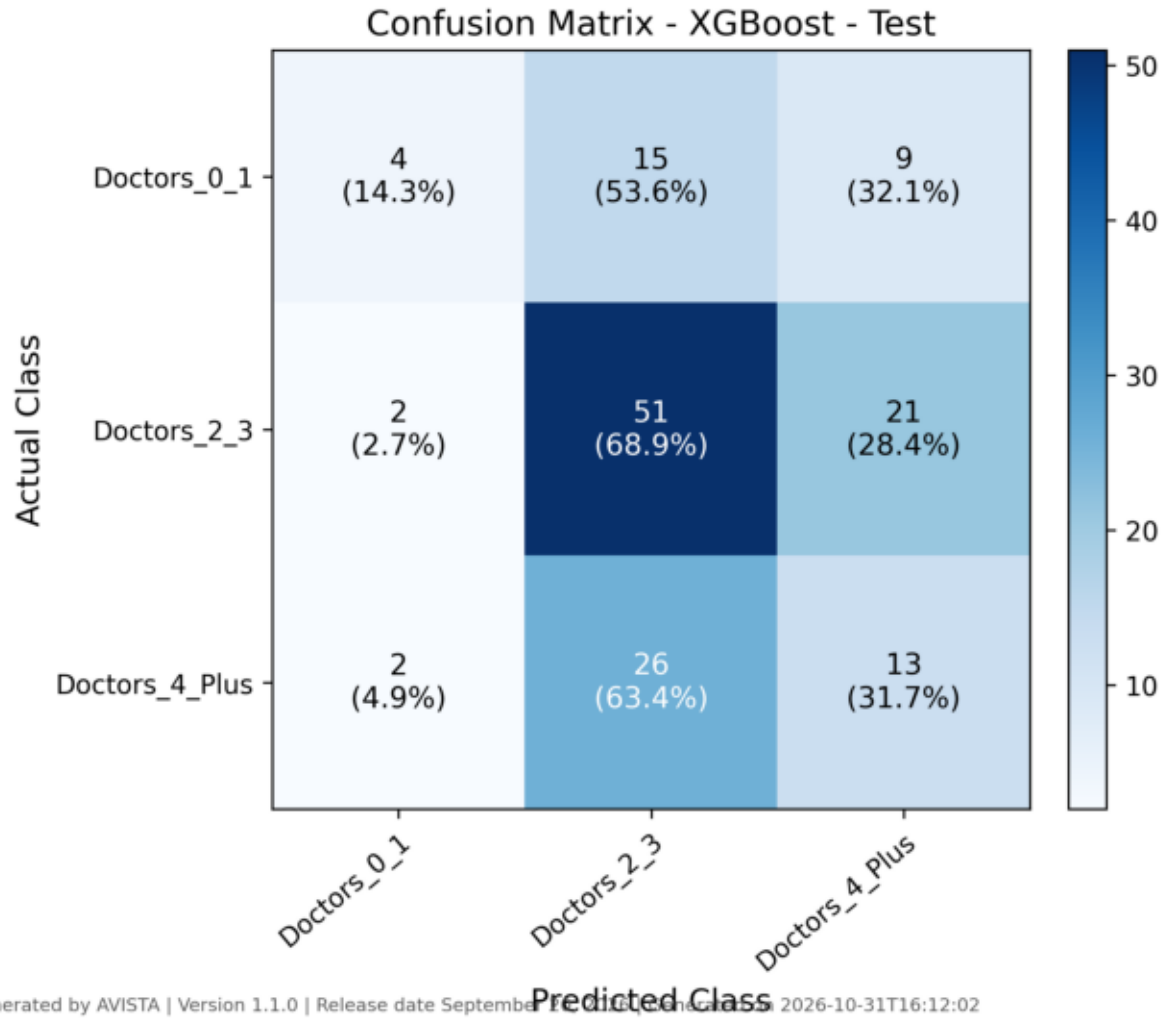
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Confusion Matrices - TabPFN 2 5



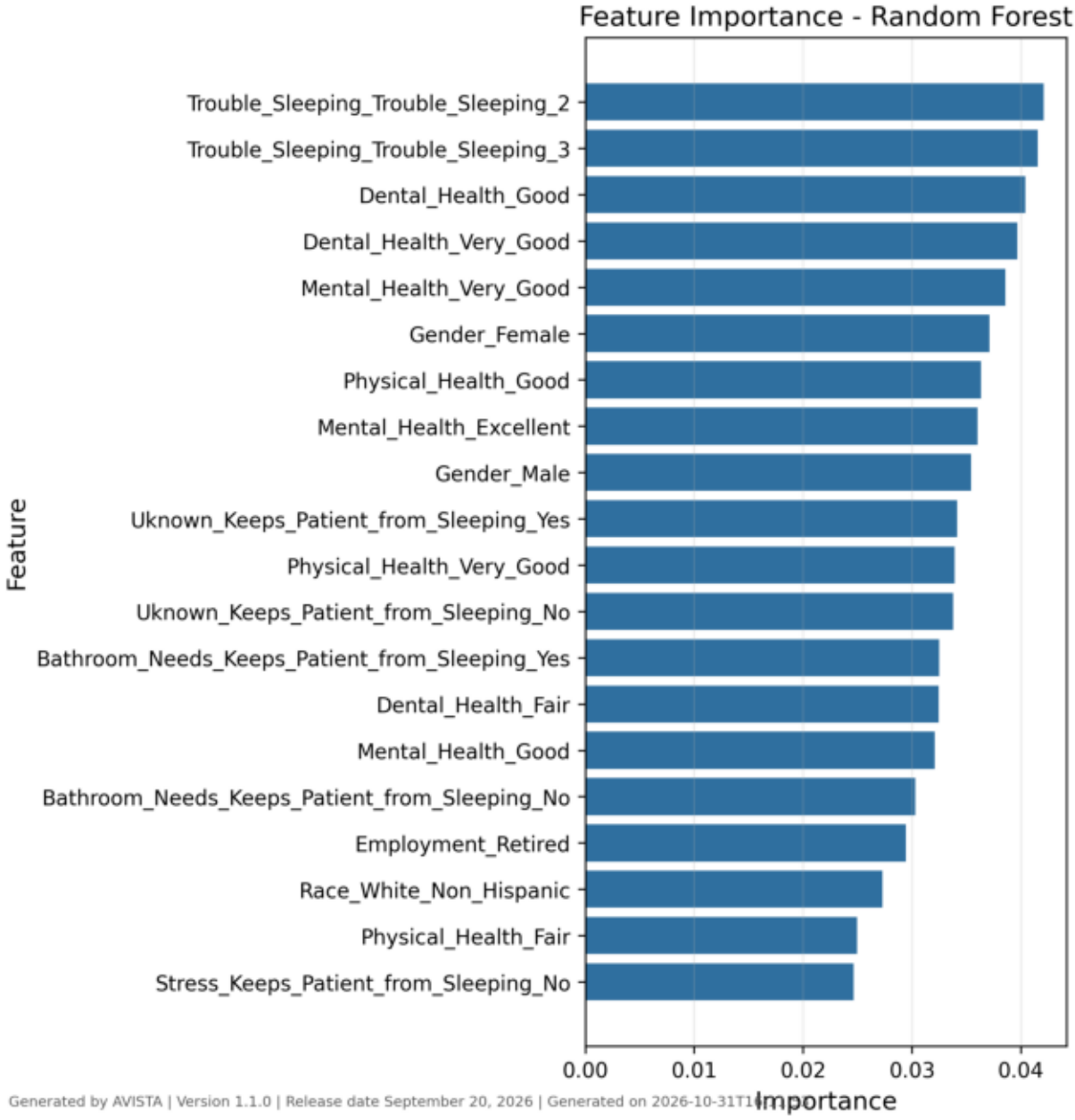
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Confusion Matrices - XGBoost



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Feature Importance - Random Forest



Test Classification Report - FT-Transformer

Class	precision	recall	f1-score	support
Doctors_0_1	0.2143	0.4286	0.2857	28.0000
Doctors_2_3	0.5556	0.2027	0.2970	74.0000
Doctors_4_Plus	0.3500	0.5122	0.4158	41.0000
accuracy	0.3357	0.3357	0.3357	0.3357
macro avg	0.3733	0.3812	0.3329	143.0000
weighted avg	0.4298	0.3357	0.3289	143.0000

Test Classification Report - Logistic Regression

Class	precision	recall	f1-score	support
Doctors_0_1	0.2424	0.2857	0.2623	28.0000
Doctors_2_3	0.5833	0.3784	0.4590	74.0000
Doctors_4_Plus	0.3387	0.5122	0.4078	41.0000
accuracy	0.3986	0.3986	0.3986	0.3986
macro avg	0.3882	0.3921	0.3764	143.0000
weighted avg	0.4464	0.3986	0.4058	143.0000

Test Classification Report - Random Forest

Class	precision	recall	f1-score	support
Doctors_0_1	0.3333	0.1786	0.2326	28.0000
Doctors_2_3	0.5556	0.6081	0.5806	74.0000
Doctors_4_Plus	0.2553	0.2927	0.2727	41.0000
accuracy	0.4336	0.4336	0.4336	0.4336
macro avg	0.3814	0.3598	0.3620	143.0000
weighted avg	0.4260	0.4336	0.4242	143.0000

Test Classification Report - TabPFN 2 5

Class	precision	recall	f1-score	support
Doctors_0_1	0.0000	0.0000	0.0000	28.0000
Doctors_2_3	0.5175	1.0000	0.6820	74.0000
Doctors_4_Plus	0.0000	0.0000	0.0000	41.0000
accuracy	0.5175	0.5175	0.5175	0.5175
macro avg	0.1725	0.3333	0.2273	143.0000
weighted avg	0.2678	0.5175	0.3529	143.0000

Test Classification Report - XGBoost

Class	precision	recall	f1-score	support
Doctors_0_1	0.5000	0.1429	0.2222	28.0000
Doctors_2_3	0.5543	0.6892	0.6145	74.0000
Doctors_4_Plus	0.3023	0.3171	0.3095	41.0000
accuracy	0.4755	0.4755	0.4755	0.4755
macro avg	0.4522	0.3830	0.3821	143.0000
weighted avg	0.4714	0.4755	0.4502	143.0000